EX No.3.2. CREATE 3D MODEL FROM ORTHOGRAPHIC PROJECTION

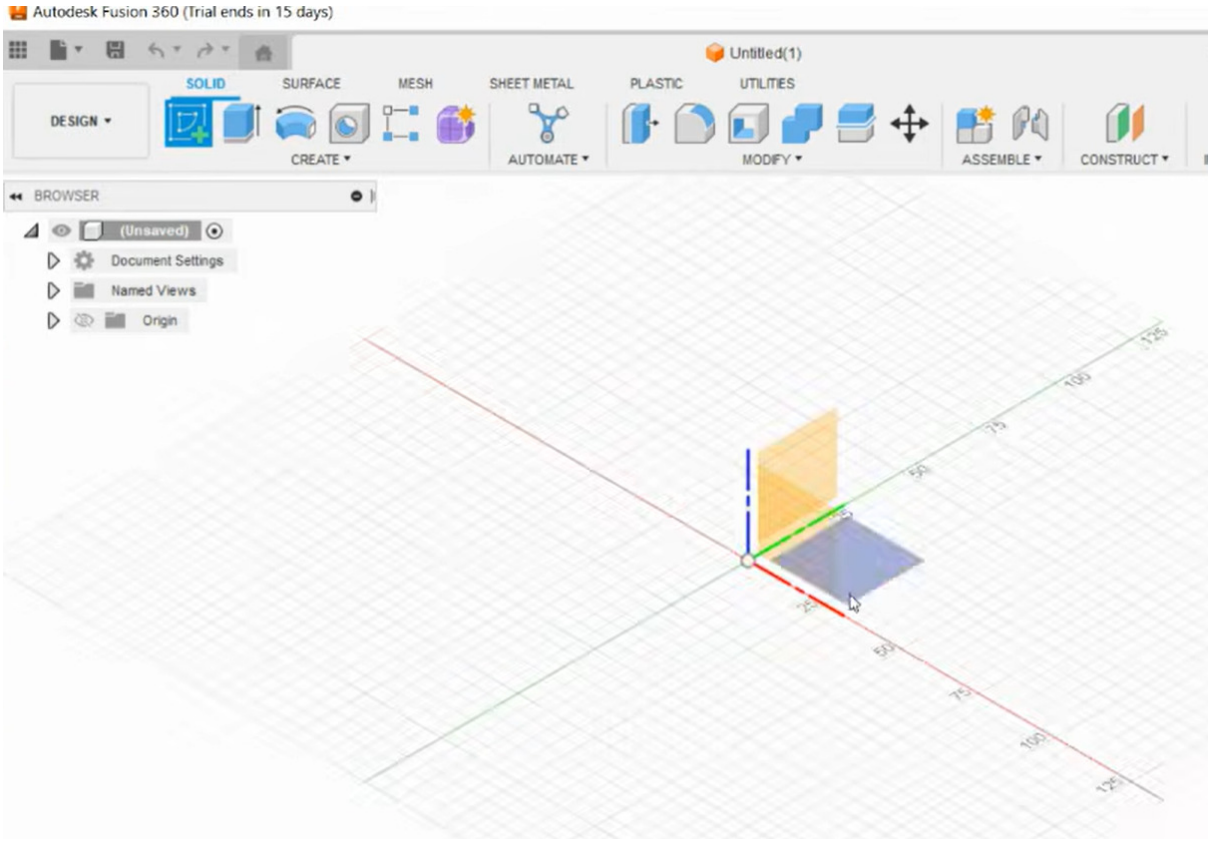
AIM:

To create a 3D model from given orthographic projection.

REQUIREMENTS:

1.Autodesk fusion 360
2.Windows 10
3.16 GB of RAM (integrated graphics recommend 1 GB or more)
4.2.5 Mbps or faster download; 500 Kbps or faster upload

PROCEDURE:
1.First we should open the fusion 360,then select new to create a new project.
2.We should select the view as front or top or which view we want.
3.Then click create sketch to start our project.

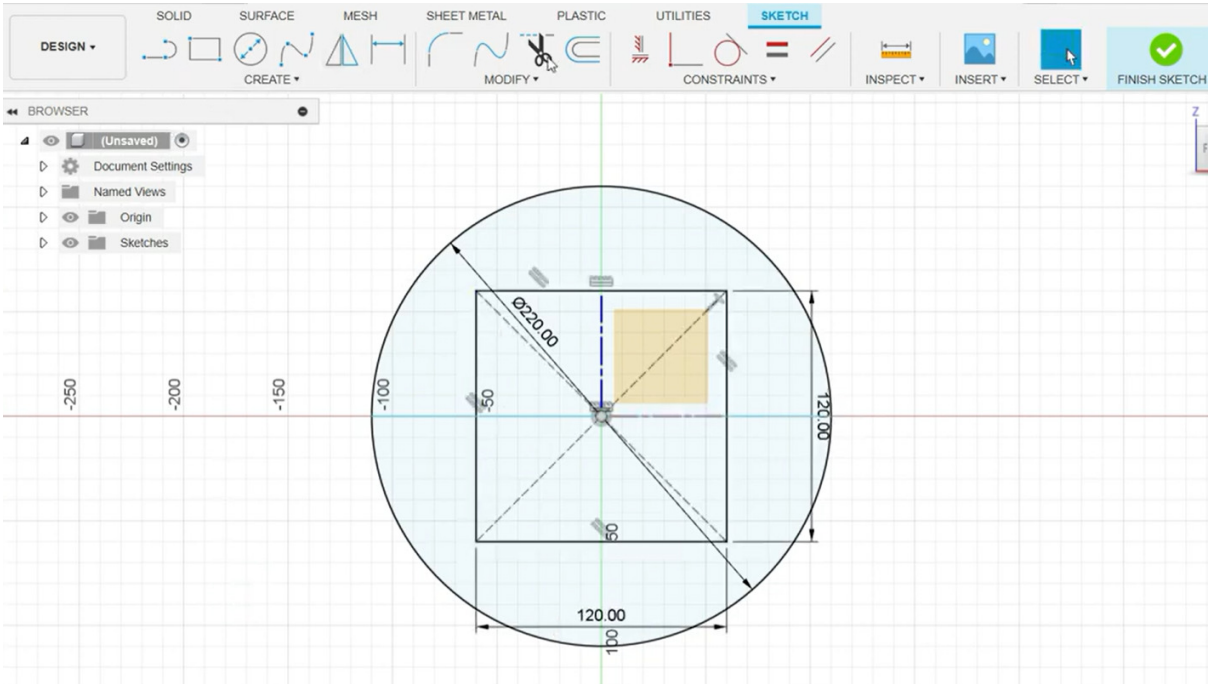


4.Draw the base rectangular portion using the Line tool according to the given dimensions.

5. Set the overall length of the base as 420 mm and the width as 150 mm using the Sketch Dimension (D) tool.

6. Draw the two side circular portions using the Arc tool and provide the radius of R60 mm.

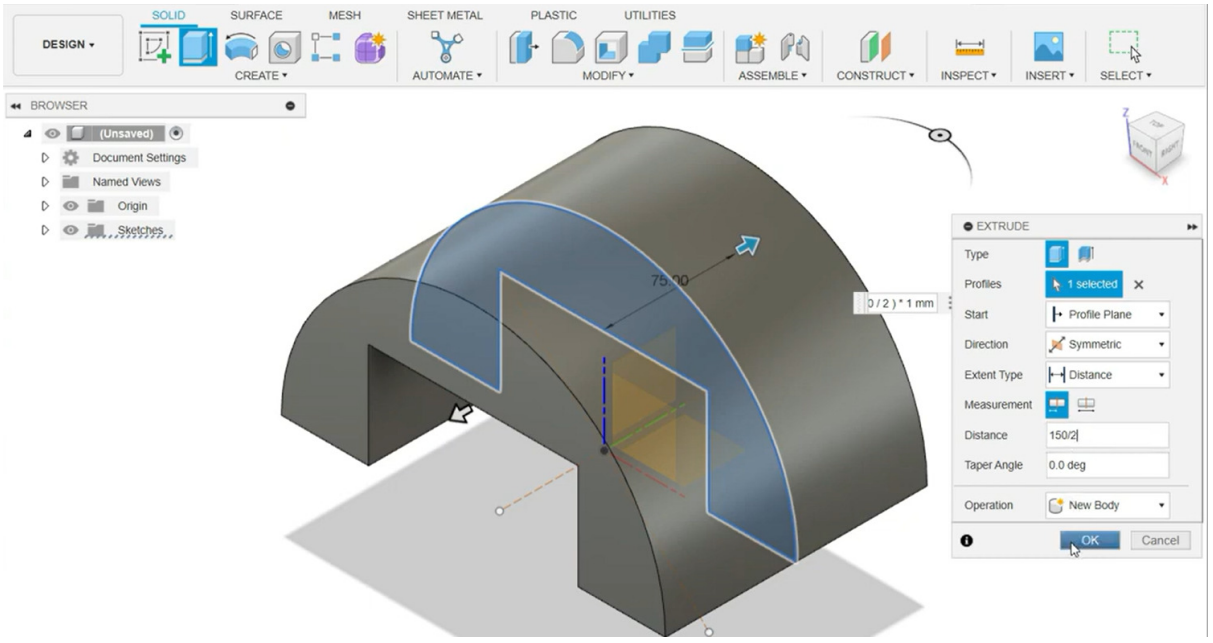
7. Use the Circle tool to create the two side holes with a diameter of Ø40 mm.



8. Draw the centre circular hole with a diameter of Ø20 mm and position it according to the given dimensions.

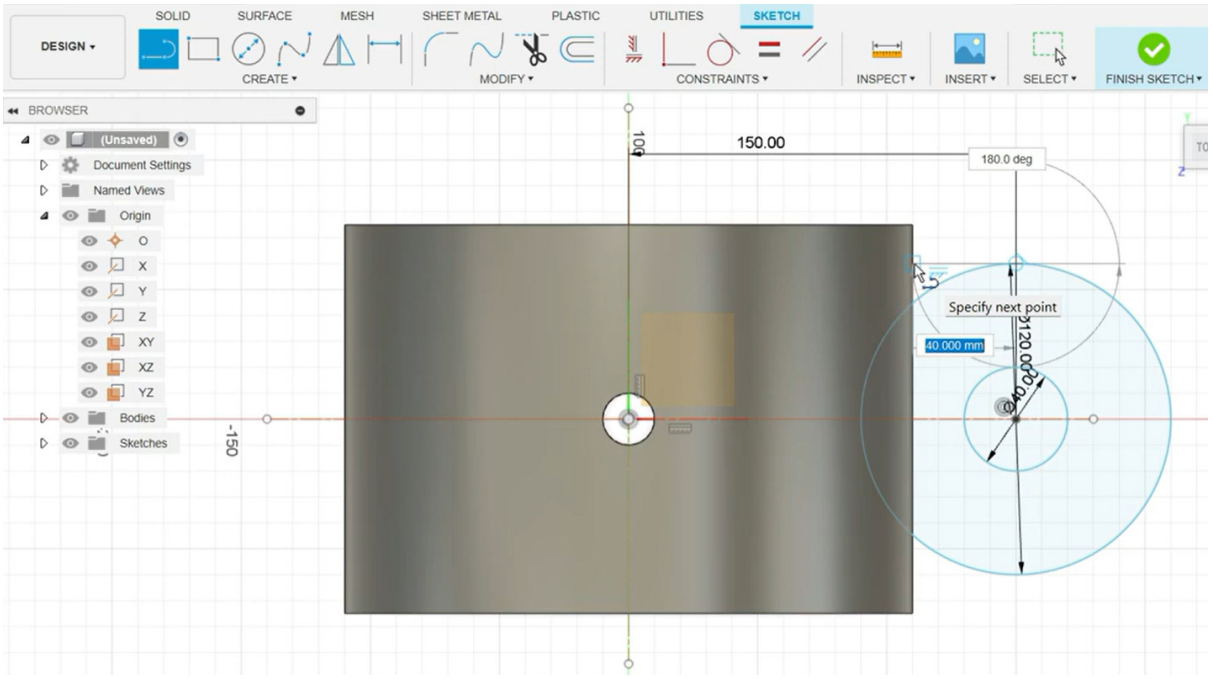
9. Create the raised curved portion at the centre using the Arc tool and set its radius as R110 mm.

10. Draw the rectangular opening at the bottom centre using the Line tool and apply the dimensions of 120 mm width and 60 mm height.



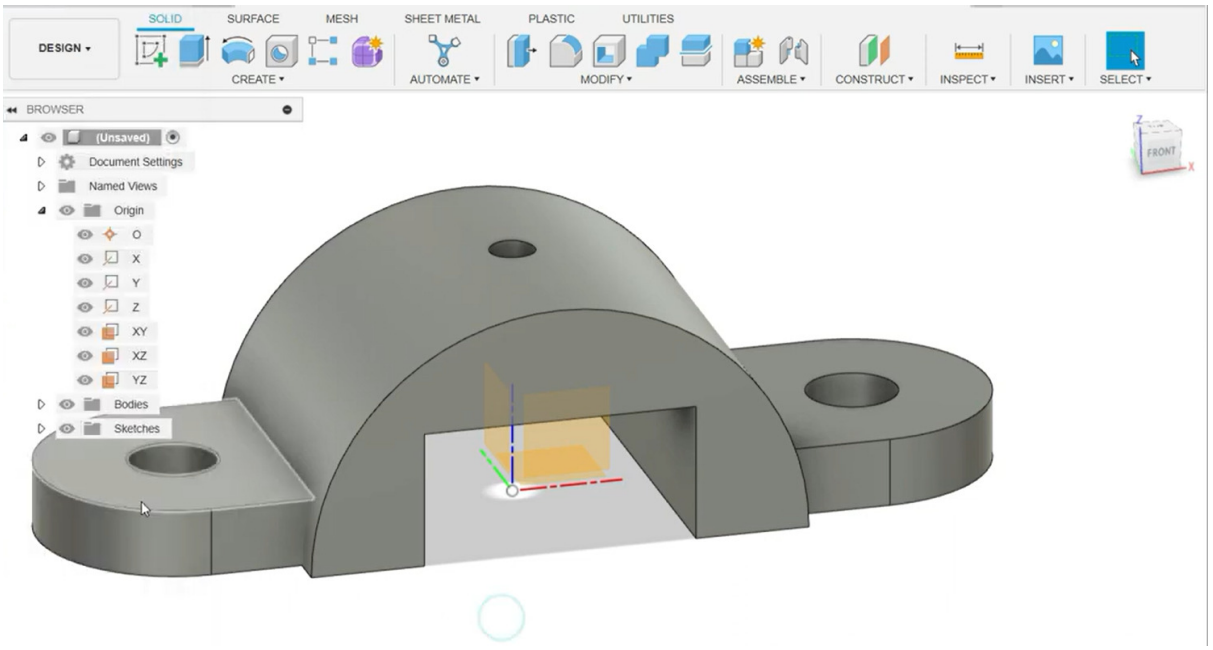
11. Complete the front profile by maintaining the base thickness of 30 mm and overall height of 110 mm.

12. Check the top view, front view and right view to ensure that all the dimensions are correctly applied.

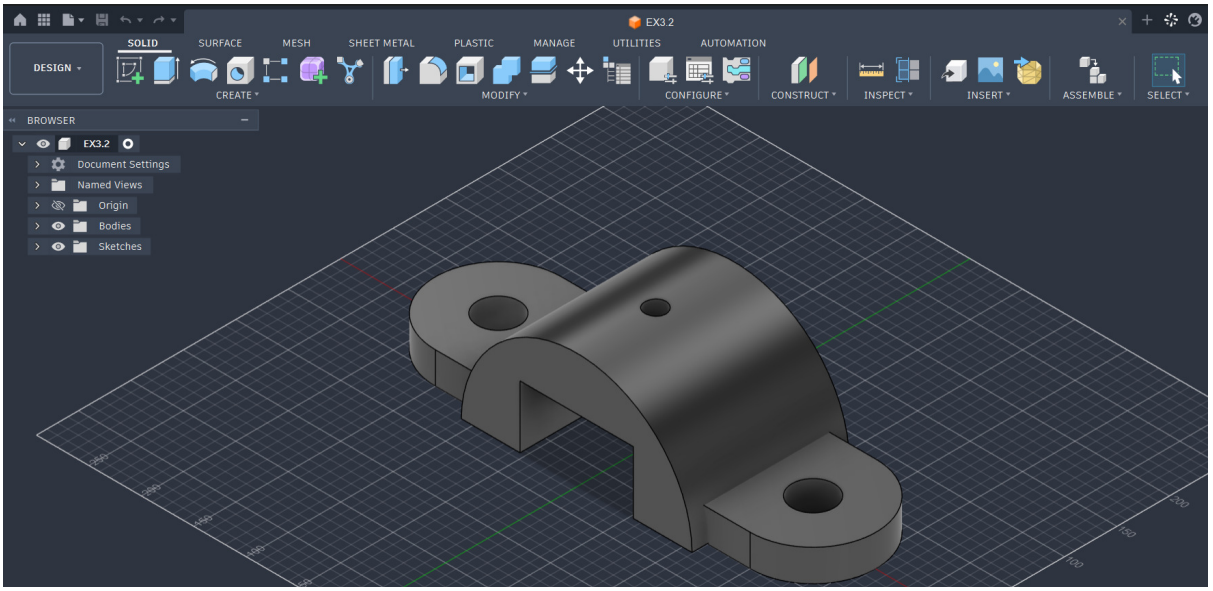


13. Finish the sketch and use the Extrude tool to provide the required width of 150 mm.

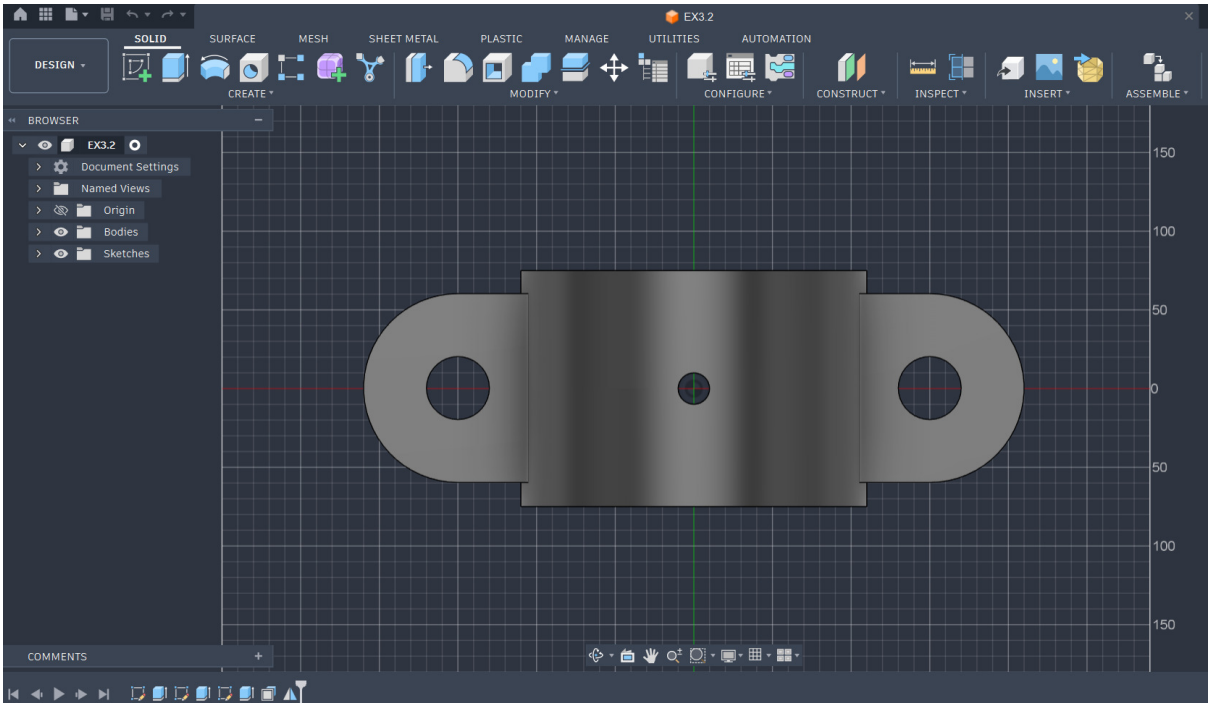
14. Use the Extrude Cut tool wherever necessary to create the required holes and openings.



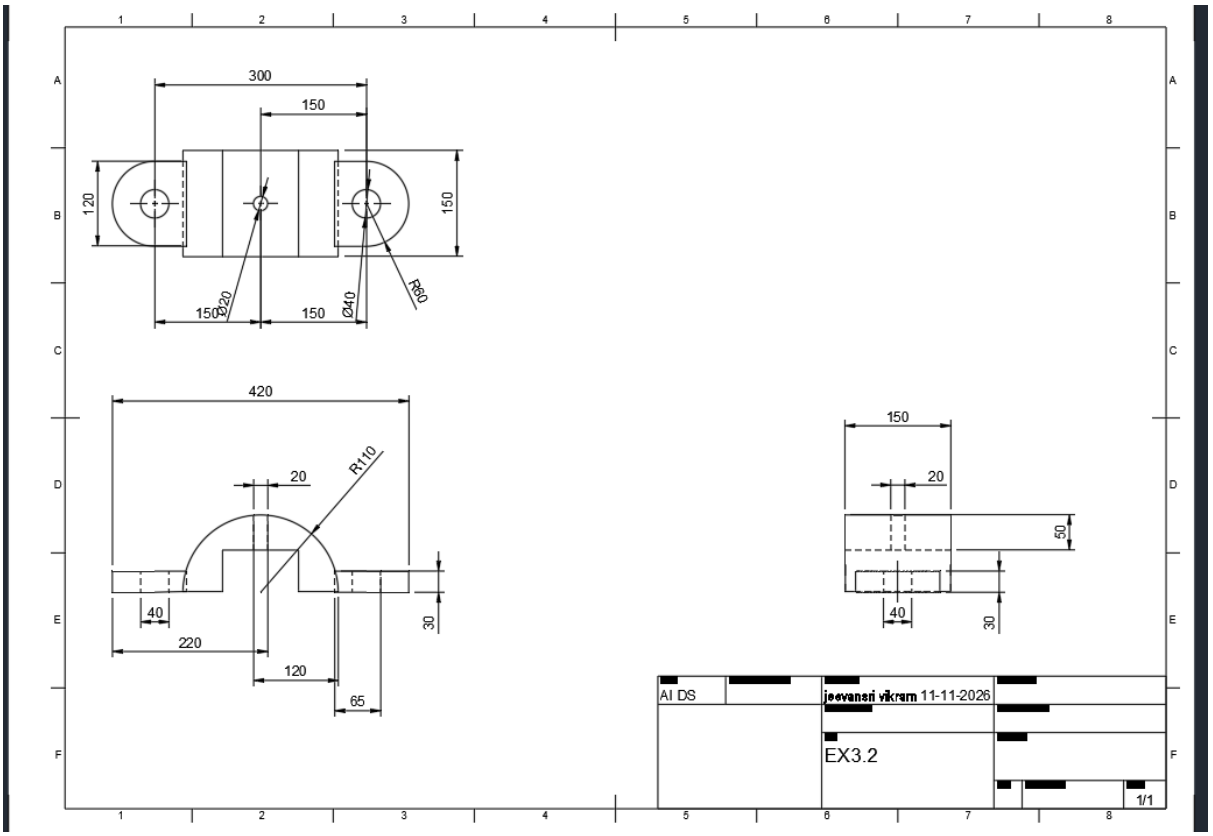
15. Check the final 3D model with the given drawing and save the completed design.



SKETCHES:



OUTPUT:



RESULT:

Thus, 3D model is created from orthographic projection.